

MAO-JAN LIN

mj.maojanlin@gmail.com ◇ maojanlin.github.io

3400 N Charles St. Baltimore, MD 21218

(667) · 444 · 9532 ◇ last updated: Sep 28, 2026

RESEARCH INTERESTS

Computational Genomics, Immunogenomics, Pangenomics, Reference Bias, Sequence Alignment, Variant Calling

EDUCATION

Johns Hopkins University

08/2021 – present

PhD Candidate in Computer Science (expected graduation: 12/2026)

Advisor: Dr. Ben Langmead

Qualifying project advisor: Dr. Yana Safonova

National Taiwan University

09/2016 – 06/2019

Master of Science in Electronics Engineering

Advisor: Dr. Yi-Chang Lu

Thesis: "A Parallel Design of Dynamic Programming Sequence Aligner with Affine Gap Traceback"

National Taiwan University

09/2012 – 06/2016

Bachelor of Science in Electrical Engineering

RESEARCH AND INDUSTRY EXPERIENCE

Johns Hopkins University

08/2021–present

Graduate Researcher, Langmead Lab

Baltimore, MD

- **ImpuT2T**: Built a Snakemake workflow that scaffolds HiFi draft assemblies and patches gaps with haplotypes from the Human Pangenome Reference Consortium (HPRC) Release 2 pangenome, outperforming single-reference patchers on recovered sequence and error rate. [[bioRxiv 2026](#); [GitHub](#)]
- **impute-first**: Developed the downstream variant-calling module for this framework, using multiple linear alignments and coordinate lifting to call variants from an imputed personalized reference. [[Genome Research](#), 2026; [GitHub](#)]
- **IGLoo**: Developed a toolkit to characterize immunoglobulin (IG) V(D)J recombination in lymphoblastoid cell lines (LCLs) and improve IG heavy chain (IGH) assemblies for HPRC samples using PacBio HiFi data. [[Cell Reports Methods](#), 2025; [GitHub](#)]
- **biastools**: Developed a framework to measure, visualize, and diagnose sequence alignment reference bias across linear and pangenome references. [[Genome Biology](#), 2024; [GitHub](#)]

Roche

06/2024–08/2024

Summer Intern, Computational Biology Molecular Lab Applications

Remote

- Developed a structural variant calling pipeline to enhance recall and precision for the Sequencing by Expansion (SBX) platform.

National Taiwan University Hospital, Lab of Dr. Pei-Lung Chen

07/2020–07/2021

Research Assistant

- Developed gAIRR Suite pipelines for adaptive immune receptor genotyping from probe-based targeted sequencing and IG/TR gene annotation in personal assemblies and reference genomes.

- Investigated the reliability of homology-based protein function annotation.
- Designed algorithms and ASIC accelerators for variant calling and sequence alignment, including memory-efficient affine-gap traceback and a CPU–ASIC hybrid workflow for BLASTn-based multiple sequence alignment.

Apple

01/2018–07/2018

Product Design Intern, Input Device Team

Taipei/Shanghai

- Built software analysis tools on the production line for Apple's upstream manufacturer.
- Developed circuit solutions and prototype software—including an embedded system with Python-based serial communication interfaces—for a new MacBook keyboard design.
- Planned experimental designs for product life test and failure analysis.

Himax

07/2017 – 08/2017

Summer Intern, Video Image Processing Team

Taipei

- Developed compression algorithms for 3D depth sensing solutions.
- Built a camera noise model based on experimental data and probability theory.

PEER-REVIEWED JOURNAL PAPERSSee [Google Scholar](#) for more information.

* Equal contribution

7. Eric Bo-Ru Chao, Siao-Jyuan Jhan, Po-Chih Lin, Yuh-Tsyh Chou, Wen-Chi Chang, Allen Chilun Luo, **Mao-Jan Lin**, Ching-Hua Kuo, Yu-An Chen, Pei-Lung Chen, Ta-Chen Su. (2026). Genetic Determinants of Chylomicronemia in Taiwan: Predominance of LPL Rare Variants and APOA5 p.G185C. *Journal of Clinical Lipidology*. <https://doi.org/10.1016/j.jacl.2026.07.005>
6. Kuan-Ta Huang, Yu-Hsuan Yang, **Mao-Jan Lin**, Yu-Hui Lin, Sheng-Kai Lai, Ting-Hsuan Chou, Chieh-Yu Lee, Tsung-Kai Hung, Chia-Lang Hsu, Ya-Chien Yang, Chien-Yu Chen, Pei-Lung Chen, Jacob Shu-Jui Hsu. (2026). gAIRR-wgs: high-resolution T cell receptor allele typing in biobank-scale whole-genome sequencing data. *Briefings in Bioinformatics* 27:bbag441. <https://doi.org/10.1093/bib/bbag441>
5. Kavya Vaddadi, **Mao-Jan Lin**, Sina Majidian, Taher Mun, Ben Langmead. (2026). Minimizing reference bias with an imputed personalized reference. *Genome Research* 36:740–753. <https://doi.org/10.1101/gr.280989.125>; [GitHub](#)
4. Yu-Hsuan Yang, Chi-Yuan Yao, **Mao-Jan Lin**, Kuan-Ta Huang, Yu-Hui Lin, I-Hsuan Chiu, Sheng-Kai Lai, Chih-Yeh Chen, Ya-Chien Yang, Chia-Lang Hsu, Jacob Shujui Hsu, Chien-Yu Chen, Pei-Lung Chen. (2026). Unmasking human T cell receptor germline diversity: 335 novel alleles identified in 47 pangenome reference individuals using the gAIRR suite. *Journal of Advanced Research*. <https://doi.org/10.1016/j.jare.2026.01.053>
3. **Mao-Jan Lin**, Ben Langmead, Yana Safonova. (2025). IGLoo enables comprehensive analysis and assembly of immunoglobulin heavy-chain loci in lymphoblastoid cell lines using PacBio high-fidelity reads. *Cell Reports Methods*. <https://doi.org/10.1016/j.crmeth.2025.101033>; [GitHub](#)
2. **Mao-Jan Lin**, Sheila Iyer, Nae-Chyun Chen, Ben Langmead. (2024). Measuring, visualizing and diagnosing reference bias with biastools. *Genome Biology* 25:101. <https://doi.org/10.1186/s13059-024-03240-8>; [GitHub](#)
1. ***Mao-Jan Lin**, *Yu-Chun Lin, *Nae-Chyun Chen, Allen Chilun Luo, Sheng-Kai Lai, Chia-Lang Hsu, Jacob Shujui Hsu, Chien-Yu Chen, Wei-Shiung Yang, Pei-Lung Chen. (2022). Profiling Genes Encoding the Adaptive Immune Receptor Repertoire with gAIRR Suite. *Frontiers in Immunology* 13:922513. <https://doi.org/10.3389/fimmu.2022.922513>; [GitHub](#)

PREPRINTS

1. **Mao-Jan Lin**, Vikram S. Shivakumar, Ben Langmead, Human Pangenome Reference Consortium. (2026). ImpuT2T: Pangenome-Based Patching for Human Genome Assemblies. *bioRxiv (preprint)*. <https://doi.org/10.64898/2026.07.27.741037>; [GitHub](#)

PEER-REVIEWED CONFERENCE PAPERS

5. Ruei-Ting Chien, **Mao-Jan Lin**, Yang-Ming Yeh, Yi-Chang Lu. (2022). Traceback Memory Reduction for Three-Sequence Alignment Algorithm with Affine Gap Models. *IEEE Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA)*. <https://doi.org/10.23919/APSIPAASC55919.2022.9980113>
4. Yu-Cheng Li, **Mao-Jan Lin**, Xiao-Xuan Huang, Chien-Yu Chen, and Yi-Chang Lu. (2020). Comprehensive Study of Keywords for Sequence-Based Automatic Annotation of Protein Functions. *IEEE Bioinformatics and BioEngineering (BIBE)*. <https://doi.org/10.1109/BIBE50027.2020.00012>
3. Ming-Hung Chen, **Mao-Jan Lin**, Yu-Cheng Li, and Yi-Chang Lu. (2019). Banded Pair-HMM Algorithm for DNA Variant Calling and Its Hardware Accelerator Design. *IEEE Bioinformatics and BioEngineering (BIBE)*. <https://doi.org/10.1109/BIBE.2019.00107>
2. **Mao-Jan Lin**, Yu-Cheng Li, and Yi-Chang Lu. (2019). Hardware Accelerator Design for Dynamic-Programming-Based Protein Sequence Alignment with Affine Gap Tracebacks. *IEEE Bio-medical Circuits and Systems Conference (BioCAS)*. <https://doi.org/10.1109/BIOCAS.2019.8919080>
1. **Mao-Jan Lin**, Chih-Yu Chang, Yu-Cheng Li, Nae-Chyun Chen, and Yi-Chang Lu. (2018). A Hybrid Flow for Multiple Sequence Alignment with a BLASTn Based Pairwise Alignment Processor. *IEEE International Symposium on Circuits and Systems (ISCAS)*. <https://doi.org/10.1109/ISCAS.2018.8351254>

PRESENTATIONS

Invited Talks

1. **National Taiwan University, Graduate School of Medical Genomics Proteomics**. *Seminar*. Taipei, Taiwan. January 8th, 2025.
Profiling immunoglobulin heavy chain in lymphoblastoid cell lines with HiFi reads.

Conference Talks

1. **Intelligent Systems for Molecular Biology (ISMB)**. Washington, D.C., USA. July 13th, 2026.
ImpuT2T: Pangenome-Based Patching for Human Genome Assemblies.

Poster Presentations

7. ImpuT2T: Pangenome-Based Patching for Human Genome Assemblies. Telomere-to-Telomere Face-to-Face (T2T-F2F). *Santa Cruz, California, USA*. September 2026.
6. Extending IGLoo to light chains: High-resolution analysis and assembly of IGK and IGL loci in lymphoblastoid cell lines with PacBio HiFi reads. Workshop on Algorithms in Bioinformatics (WABI). *College Park, Maryland, USA*. 2025.
5. IGLoo—Profiling the immunoglobulin heavy chain locus in lymphoblastoid cell lines with PacBio high-fidelity sequencing reads. Cold Spring Harbor Laboratory Meeting on Biological Data Science. *Cold Spring Harbor, New York, USA*. 2024.
4. Measuring, visualizing and diagnosing reference bias with biastools. Cold Spring Harbor Laboratory on Genome Informatics. *Cold Spring Harbor, New York, USA*. 2023.

3. Biastools provides detailed measurements and diagnoses for reference bias. Workshop on emerging methods in sequence analysis (WEMSA). *University Park, Pennsylvania, USA*. 2023.
2. Profiling germline adaptive immune receptor repertoire with gAIRR Suite. Cold Spring Harbor Laboratory on Genome Informatics. *Virtual*. 2021.
1. A Hybrid Flow for Multiple Sequence Alignment with a BLASTn Based Pairwise Alignment Processor. IEEE International Symposium on Circuits and Systems (ISCAS). *Florence, Italy*. 2018.

TEACHING AND MENTORSHIP

TEACHING EXPERIENCE

- **Department of Computer Science, Johns Hopkins University**
Teaching Assistant
 EN.601.446/646 Sketching and Indexing, Spring 2025
 EN.601.451 Introduction to Computational Immunogenomics, Fall 2022
Guest Lecturer
 EN.601.451 Introduction to Computational Immunogenomics, Fall 2022
- **Department of Biomechanical Engineering, National Taiwan University**
Teaching Assistant and Guest Lecturer
 BME5938 Bioinformatics Algorithms, Summer 2020

MENTORSHIP

- **Department of Computer Science, Johns Hopkins University**
 Ethan Chang, undergraduate researcher (2025)
 Mentored an undergraduate researcher in designing and benchmarking experiments to assess how alignment accuracy changes with pangenome size and identify diminishing returns from additional genomes.
- **Department of Electrical Engineering, National Taiwan University**
 Ming-Hung Chen, undergraduate researcher (2019)
 Mentored an undergraduate researcher in designing and implementing a hardware accelerator for GATK HaplotypeCaller.

SOFTWARE

- **ImpuT2T** - Pangenome-based patching for draft assemblies.
<https://github.com/maojanlin/ImpuT2T>
- **impute-first** - Personalized reference building through imputation to improve variant calling.
 Developed in collaboration with Kavya Vaddadi and Sina Majidian.
<https://github.com/kvaddadi/impute-first>
- **IGLoo** - Measures V(D)J recombination in IGH loci from human LCLs.
<https://github.com/maojanlin/IGLoo>
- **biastools** - Measures reference bias in sequence alignment across different setups.
<https://github.com/maojanlin/biastools>
- **gAIRR Suite** - Profiles human T-cell receptor (TR) and IG genes from sequence data.
<https://github.com/maojanlin/gAIRRsuite>

SERVICE

Peer Review Experience

2026– BMC Bioinformatics (sub-reviewer)

2026– ISMB (sub-reviewer)

2026– RECOMB (sub-reviewer)

Service

2023– Peer mentorship program - Department of Computer Science, Johns Hopkins University